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Immunomodulatory effects of aqueous extract of *Tridax procumbens* in experimental animals

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Abstract

The immunomodulatory properties of ethanol insoluble fraction of aqueous extract of *Tridax procumbens* Linn. (TPEIF) have been investigated. After intraperitoneal administration of TPEIF in doses of 0.25 and 0.5 g/kg body weight (BW) a significant increase in phagocytic index, leukocyte count and spleenic antibody secreting cells was noticed. Stimulation of humoral immune response was further observed with elevation in heamagglutination antibody titer. Heightened delayed type hypersensitivity reaction suggested convincing evidence for activation of cellular immune system. Protective action of herbal medicine in case of anaphylactic shock was also studied. In addition, elicitation of specific antibody titer against tetanus toxoid (TT) challenge was measured in order to explore the possible use as adjuvant along with clinical vaccination program to reduce number of non-responders. The results suggest that TPEIF influences both humoral as well as cell mediated immune system vis-a-vis assists in genesis of improved antibody response against specific clinical antigen.

Keywords: Immunomodulation; *Tridax procumbens*; Humoral immunity; Cellular immunity; Delayed type hypersensitivity; Herbal immunomodulator

1. Introduction

Herbal medicine has become an integral part of standard healthcare, based on a combination of time honoured traditional usage and ongoing scientific research. Burgeoning interest in medicinal herbs has increased scientific scrutiny of their therapeutic potential and safety. Some of the medicinal plants are believed to enhance the natural resistance of the body to infections (Atal et al., 1986). The modulation of immune response with the aid of various bioactives in order to alleviate certain diseases is an active area of interest. Apart from being specifically stimulatory or suppressive, certain agents normalize or modulate pathophysiological processes and are hence called ‘immunomodulatory agents’ (Wagner, 1983). The property of any substance to enhance non-specific resistance of body against pathogens is termed ‘adaptogenic.’ Most important area in which herbal medicine has not to witness any breakthrough is the development of adjuvants to be used in vaccination programs or immunosuppressants that can be safely exploited in organ transplantation and autoimmune diseases. These fundamental fields of immunomodulators are currently receiving inadequate attention. A number of plant products are being investigated for immune response modifying activity (Upadhyay, 1997). A plethora of plant-derived materials (proteins, lectins, polysaccharides, etc.) have been shown to stimulate the immune system (Tzianabos, 2000). Some of the plants with established immunomodulatory activity are *Viscum album*, *Panax ginseng*, *Asparagus racemosus*, *Azadirachta indica*, *Tinospora cordifolia*, *Polygala senega*, *Ocimum santum* (SaiRam et al., 1997; Estrada et al., 2000; Mediratta et al., 2002).

Tridax procumbens Linn. (TP) family Compositae commonly known as ‘Ghamra’ and in English popularly called ‘coat buttons’ because of appearance of flowers has been extensively used in Ayurvadic system of medicine for various ailments and is shown to posses significant antiinflammatory, hepatoprotective, wound healing and antimicrobial properties (Diwan et al., 1989; Pathak et al., 1991; Saraf et al., 1991; Udupa et al., 1991; Perumal et al., 1999; Taddei and Rosas, 2000). The present study is aimed at studying the immunomodulatory acitivity of aqueous extract of TP in mice.

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2. Materials and methods

2.1. Reagents

Agarose and bovine serum albumin (BSA) were purchased from Himedia, India. Tetanus toxoid (TT) was obtained from Biological E, India. Tween 20 and ovalbumin (OV) were purchased from Sigma, USA. Goat anti-mice IgG-HRP conjugate and tetramethyl benzidine (TMB) were procured from Genei, Banglore, India. All other reagents were of analytical grade unless otherwise mentioned.

2.2. Plant material and extraction

The aerial parts of TP were collected in the month of July and August from the botanical garden of Dr. Harisingh Gour University and dried under shade at room temperature. Dried material was coarsely pulverized to powdered form. One kilogram of powdered plant material was extracted using a Soxhlet apparatus with 500 ml distilled water at 60–70 °C for 35 complete cycles. The aqueous extract was dried at 30–40 °C using a vacuum evaporator and the resulting powder was fractioned in to ethanol soluble and ethanol insoluble fractions. *Tridax procumbens* ethanol insoluble fraction (TPEIF) of the aqueous extract was investigated for immunomodulatory activity. The dried TPEIF was dissolved in PBS (pH 7.4) and filtered through 0.22 μm membrane filter (Millipore, USA).

2.3. Experimental animals

Swiss male albino mice weighing 25–30 g were used. They were fed a standard pellet diet and tap water ad libitum in animal house facility and maintained under standard laboratory conditions. All protocols were approved by the ethical committee of the university instituted for animal handling.

2.4. Treatment of mice

Two groups of mice each consisting of five animals received TPEIF intraperitonially (i.p.) in doses of 0.25 and 0.5 g/kg body weight (BW) for 5 consecutive days. The control group was treated with 0.5 ml phosphate buffer saline (PBS, pH 7.4), whereas native group was devoid of any treatment. No mortality or any toxic effects were observed in the above mentioned doses of TPEIF administered i.p.

2.5. Phagocytic activity

TPEIF was injected intraperitoneally (0.25 and 0.5 g/kg BW) for 5 consecutive days. Control was given equal volume of phosphate buffer saline (pH 7.4). After 48 h of last dose mice were injected via the tail vein with colloidal carbon (Indian ink), which was diluted with PBS (pH 7.4) to eight times before use (10 μl/g BW). Blood samples were drawn from the retro-orbital plexus at 0, 3, 6, 9, 12 and 15 min. The blood (25 μl) was dissolved in 0.1% sodium carbonate (2 ml) and the absorbance was measured at 660 nm (Gonda et al., 1990). The phagocytic index, K, was calculated by equation:

$$K = \frac{\ln \mathrm{OD}_1 - \ln \mathrm{OD}_2}{t_2 - t_1}$$

where, OD_1 and OD_2 depict the optical densities at times t_1 and t_2, respectively.

2.6. Leukocyte count

Two groups of mice were treated with different doses of TPEIF for 5 days. On day 6, blood was collected from retro-orbital plexus for white blood cells (WBC) count. The animals were sacrificed by cervical dislocation and their spleens were harvested for weighing and spleen leukocytes count (Puri et al., 1992). The results of these analyses were compared with that of control and native groups.

2.7. Haemagglutination antibody titer

Animals in two different groups were injected i.p. 0.2 ml of 5×10^9 SRBC on day 0. TPEIF was administered i.p. 0.25 and 0.5 g/kg BW to animals on −4, −2, 0, 2, 4 days. Control group received equal volume of PBS (pH 7.4). Blood samples were collected from retro-orbital plexus on day 7. Antibody titer was determined following the procedure reported by Nelson and Mildenhall, 1967. To two-fold dilutions of serum samples made in 25 μl volumes of normal saline containing 0.1% BSA (BSA saline) in V bottom heamagglutination plates (Tarsons, India) were added 25 μl of 0.1% suspension of SRBC in BSA saline. After thorough mixing SRBC were allowed to settle at room temperature for 90 min until control wells showed small button of cells (negative pattern). The values of the highest serum dilution causing visible haemagglutination was considered as the antibody titer.

2.8. Plaque forming cell (PFC) assay

The assay was performed according to the method reported by Jerne and Nordin, 1963. Briefly, the spleen cells of SRBC immunized TPEIF treated mice were separated in RPMI-1640 medium, washed twice and suspended in same medium (10^6 cells/ml). Glass petridishes were layered with 1.2% agarose in 0.15 M NaCl to form bottom layer. A mixture of 2 ml agarose (0.6%) in RPMI-1640 medium, 0.1 ml suspension of 20% SRBC and 1×10^6 spleen cells in a volume of 0.1 ml was poured over the bottom layer of agarose followed by an incubation period of 90 min at 37 °C. A 2 ml quantity of 1:9 diluted fresh rabbit serum was added to petridish and plate was reincubated for 40 min to allow the formation of plaques. The number of plaques was counted immediately and values are expressed as counts per 10^6 spleen cells.

2.9. Delayed type hypersensitivity (DTH)

Hypersensitivity reaction to SRBC was induced in mice following the method reported by Ray et al., 1996. Animals were sensitized with 10% SRBC (1×10^8 cells) at day 0 and day 7 subcutaneously (s.c.). They were divided into two groups: one group was treated with TPEIF 0.25 g/kg BW i.p. and other received 0.5 g/kg BW TPEIF i.p. on days −4, −2, 0, 2, 4, 6, 8, whereas control group was administered with equal volume of PBS (pH 7.4). On day 9, both groups were challenged with 1×10^8 SRBC cells, intradermally into the left footpad of each mouse, while PBS (pH 7.4) was injected into right hind paw. The increase in footpad thickness (FPT) was measured 24 h after SRBC challenge by volume differential meter. The degree of DTH reaction was expressed as the percentage increase in FPT over the control values.

2.10. Systemic anaphylaxis reaction

Animals were sensitized by 1 mg bovine serum albumin subcutaneously in 0.2 ml aluminum hydroxide (15 mg/ml) at day 0 and were shocked by intravenous injection of 1 mg BSA in PBS (pH 7.4) on day 14. 1 mg of ovalbumin i.v. in 0.2 ml PBS (pH 7.4) served as negative control. Immunomodulatory activity was examined by injecting 0.25 and 0.5 g/kg BW of mice on alternative days from day 0 to day 14 for seven times before shocking injection. The systemic anaphylactic reaction was observed within 30 min following the shocking injection and rated in following fashion: positive reaction, mortality or animal rendered still for at least 1 min; negative reaction with no change or normal movement (Hsu et al., 1997).

2.11. Antibody response against specific antigen

Conventional marketed alum adsorbed tetanus toxoid was used to generate specific antibody titer. Priming was done on day 0 with booster dose on day 28. TPEIF was administered i.p. in two doses of 0.25 and 0.5 g/kg BW on day −4, −2, 0, 2, 4, 6, 8. Control group received PBS (pH 7.4) in place of extract, while native group was devoid of any treatment. TT challenge was given to all groups except native animals. Blood sampling was done on day 14 and day 42 to measure antibody titer against TT. Anti-TT antibodies in mice sera were estimated using ELISA protocol (Raghuvanshi et al., 2001). Briefly, 1 μg of TT in 100 μl of PBS (pH 7.4) was coated in each well of 96 well flat bottom polystyrene immunoplate (Tarsons). The plates were washed with PBS-T (0.5% Tween 20 in PBS) and different dilutions of test serum in PBS-T were added to the wells and incubated for 1 h at room temperature. After washing with PBS-T for three times, goat-anti mice antibody-horseradish peroxidase conjugate was added and incubated for 1 h at room temperature followed by addition of tetramethyl benzidine. After 30 min reaction was stopped with 2 M H_2SO_4 and absorbance was measured at 450 nm using Labsystems, Finland microplate reader. Simultaneously, standard curve from known quantities of anti-TT antibodies was also prepared.

2.12. Statistical analysis

Data were expressed as the mean ± standard deviation (S.D.) and statistical analysis was performed using Student's *t*-test.

3. Results

3.1. Phagocytic activity

Carbon clearance test was conducted to establish phagocytic activity of reticuloendothelial system (RES) after treatment with TPEIF (Fig. 1). Phagocytic index was significantly increased after the administration of TPEIF compared to control group ($P < 0.05$). Relatively less difference was recorded between two doses of ethanol insoluble fraction, with higher dose (0.5 g/kg BW) showing phagocytic index value of 0.46 ± 0.04 compared to 0.32 ± 0.06 recorded with the dose of 0.25 g/kg BW.

3.2. Leukocyte count

White blood cell count was increased with the treatment of TPEIF compared to control and native groups. In addition, a positive effect on spleen weight and spleen leukocyte count was also observed (Table 1). However, dose response was not proportionate, i.e. with double the dose of TPEIF proportionate shoot up in response was not witnessed.

3.3. Heamagglutination antibody titer

A dose related increase in heamagglutination antibody titer was observed with five doses of TPEIF (Table 2) compared to control group. However, the response was not doubled on increasing doses of TPEIF.

3.4. Plaque forming cell assay

TPEIF administration for 5 days beginning from the day of sensitization with SRBC produced 30–35% increase in antibody secreting cells in mouse spleen (Table 2). Effect was significant compared to control group ($P < 0.01$) however, higher dose did not produced corresponding elevation in antibody secreting cells.

3.5. Delayed type hypersensitivity reaction

The effect of TPEIF administration on T-cell mediated DTH reaction is shown in Table 3. A significant

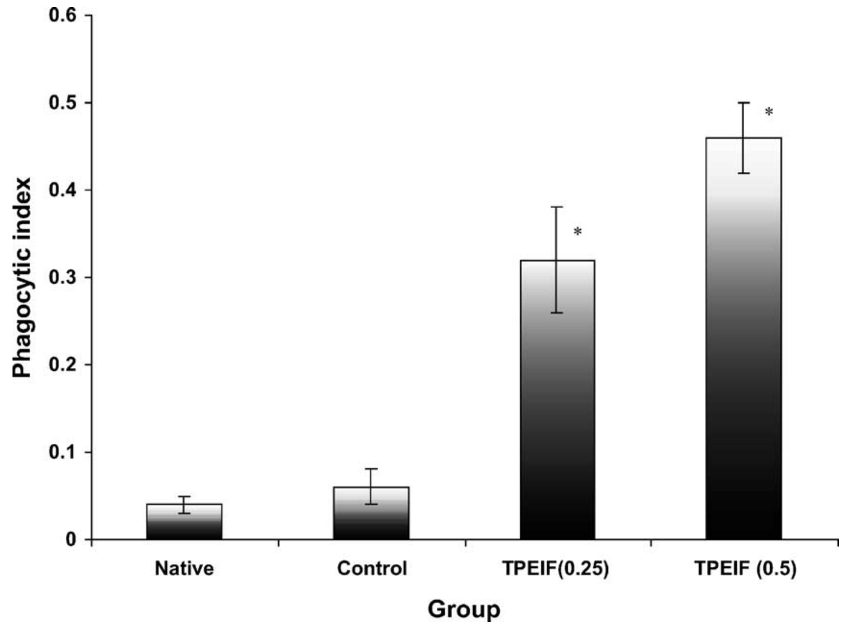

Fig. 1. Effect of TPEIF treatment on carbon clearance index $^{*}P < 0.05$ (compared to control group).

Table 1
Effect of TPEIF on spleen weight, WBC count and spleen leukocyte count

Group (treatment, TPEIF)	Spleen weight (g)	WBC (count/mm^3)	Spleen leukocyte count
Native (no treatment)	0.126 ± 0.032	1520 ± 623	30160 ± 8263
Control (PBS)	0.142 ± 0.046	2140 ± 736	51240 ± 9632
0.25 g/kg BW	0.221 ± 0.036*	3482 ± 764*	132670 ± 12276**
0.50 g/kg BW	0.235 ± 0.042*	3516 ± 812*	136420 ± 13624**

Values are expressed as mean ± S.D.
* $P < 0.05$ (compared to respective control).
** $P < 0.01$ (compared to respective control).

Table 2
Effect of TPEIF treatment on plaque forming cells and heamagglutination antibody titer

Group (treatment, TPEIF)	Heamagglutination antibody titer	Plaque forming cells/10^6 spleen cells
Control (PBS)	52.4 ± 14.2	860 ± 3.3
0.25 g/kg BW	180.3 ± 34.7*	1230 ± 4.1**
0.50 g/kg BW	194.2 ± 41.6*	1310 ± 4.7**

Values are expressed as mean ± S.D.
* $P < 0.05$ (compared to respective control).
** $P < 0.01$ (compared to respective control).

Table 3
Effect of TPEIF on delayed type hypersensitivity footpad thickness

Group (treatment, TPEIF)	Percentage increase in paw volume
Control (PBS)	21.27 ± 2.36
0.25 g/kg BW	29.42 ± 3.14*
0.50 g/kg BW	31.63 ± 3.28*

Values are expressed as mean ± S.D.
* $P < 0.05$ (compared to respective control).

enhancement (8–10%) in foot pad thickness of TPEIF treated mice was observed when compared to control group ($P < 0.05$).

3.6. Active systemic anaphylaxis

Effect of TPEIF treatment on active systemic anaphylaxis is shown in Table 4. Results indicate a positive effect on anaphylaxis with the treatment of TPEIF suggesting a reduction in number of animals presenting anaphylactic

Table 4
Effect of TPEIF on active systemic anaphylaxis

Sensitizing injection	Shocking injection	TPEIF treatment	Number of mice		
			AS	D	T
BSA (s.c.)	BSA (i.v.)	–	8	2	8
BSA (s.c.)	OV (i.v.)	–	0	0	8
BSA (s.c.)	BSA (i.v.)	0.25 g/kg BW	6	0	8*
BSA (s.c.)	BSA (i.v.)	0.50 g/kg BW	4	0	8*

Values are expressed as mean ± S.D. $^{*}P < 0.01$ (compared to positive control group). Abbreviations: BSA: bovine serum albumin; OV: ovalbumin; s.c.: subcutaneous; i.v.: intravenous; AS: anaphylactic symptoms; D: deaths; T: total mice.

symptoms with no death observed. At the other extreme, in the case of positive control group with no TPEIF treatment, all mice exhibited anaphylactic symptoms with two deaths, whereas negative control group showed no symptoms of anaphylaxis.

3.7. Specific antibody titer

Anti-TT antibody IgG titer is shown in Fig. 2. Control group exhibited an increased antibody titer with priming as well as booster dose of TT. Treatment of TPEIF further enhanced the effect of TT injection with a corresponding elevation in anti-TT IgG antibodies, however, level of antibodies was not increased proportionately with dose.

4. Discussion and conclusions

The prime objective of the study was to investigate the immunomodulatory effect of *Tridax procumbens* Linn. (TP). There is a growing interest in identifying herbal immunomodulators ever since their possible use in modern medicine has been suggested (Atal et al., 1986; Sharma et al., 1994; Lee et al., 1995). The findings outlined above have demonstrated that TPEIF possesses potent immunostimulant action.

Phagocytosis represents an important innate defence mechanism against ingested particulates including whole pathogenic microorganisms. The specialized cells that are capable of phagocytosis include blood monocytes, neutrophils and tissue macrophages. Once particulate material is ingested into phagosomes, the phagosomes fuse with lysosomes and the ingested material is then digested. Enhanced uptake of particulate matter with the treatment of TPEIF is evident from carbon clearance test. However, a dose proportionate response was not observed, since immune response is not always directly related with the immunomodulator concentration. This may be partly due to different constituents present in fraction at different concentrations and saturation of cells responsible for immune response. Some of these constituents may have immunosuppressive activity, whereas other possesses immunostimulant action (Rezaeipoor et al., 2000). The resultant effect of TPEIF is immunostimulation with a typical plateau effect beyond the dose of 0.25 g/kg BW. A variety of white blood cells, or leukocytes, including neutrophils, lymphocytes, monocytes, eosinophils and basophils participate in the development of an immune response. Of these cells, only the lymphocytes possess the attributes of diversity, specificity, memory and self/non-self recognition, the hallmark of an immune response. All the other cells play accessory roles, serving to activate lymphocytes, to increase the effectiveness of antigen clearance by phagocytosis or to secrete various immune effector molecules. Among different organs of immune system, spleen represents a major secondary lymphoid organ involved in elicitation of immune response. Unlike lymph nodes, which are specialized to trap-localized antigen from regional tissue spaces, the spleen is adapted to filtering blood and trapping blood-borne antigens and thus can respond to systemic infections. Results revealed an increase in the blood leukocytes count, weight of spleen and spleenic leukocytes count suggesting an uplift of immune status. Number of antibody secreting cells from spleen was determined using plaque forming cell assay. Since spleen contributes immensely to the humoral as well as cellular

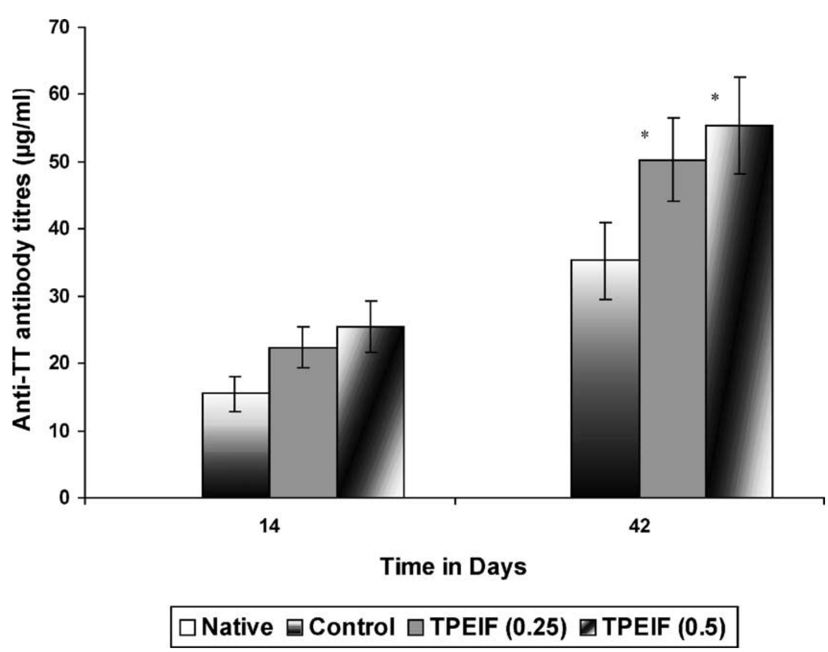

Fig. 2. Specific antibody titer against TT booster dose on 28 day $^{*}P < 0.05$ (compared to control group).

arms of immune system, its role in generation of antibody secreting cells was studied. Results suggested an enhancement in number of cells secreting antibodies against SRBC, which served as specific antigen.

Haemagglutination antibody titer was determined to establish the humoral response against SRBC. At neutral pH, red blood cells possess negative ions cloud that makes the cells repel from one another, this repulsive force is referred to as zeta potential. Because of its size and pentameric nature, IgM can overcome the electric barrier and get cross-link red blood cells, leading to subsequent agglutination. The smaller size and bivalency of IgG, however, makes them less capable to overcome the electric barrier. This characteristic may accounted for, IgM being more effective than IgG in agglutinating red blood cells (Kuby, 1994). TPEIF treatment improved the heamagglutination antibody titer reflecting an overall elevation of humoral immune response.

Delayed type hypersensitivity reaction is characterized by large influxes of non-specific inflammatory cells, in which the macrophage is a major participant. It is a type IV hypersensitivity reaction that develops when antigen activates sensitized T_{DTH} cells. These cells generally appear to be a T_H1 subpopulation although sometimes T_C cells are also involved. Activation of T_{DTH} cells by antigen presented through appropriate antigen presenting cells results in the secretion of various cytokines including interleukin-2, interferon-γ, macrophage migration inhibition factor and tumor necrosis factor-β (Askenase and Van Loveren, 1983). The overall effects of these cytokines are to recruits macrophages into the area and activate them, promoting increased phagocytic activity vis-a-vis increased concentration of lytic enzymes for more effective killing. Several lines of evidence suggest that DTH reaction is important in host defense against parasites and bacteria that can live and proliferate intracellularly. Treatment of TPEIF enhanced DTH reaction, which is reflected from the increased footpad thickness compared to control group suggesting heightened infiltration of macrophages to the inflammatory site. This study may be supporting a possible role of TPEIF in assisting cell-mediated immune response. The sesquiterpene lactones, which are isolated from the Tridax, have been identified and are known to cause delayed type hypersensitivity (Picman, 1986). So it might be possible that these sesquiterpene lactones may be present in the TPEIF.

Anaphylatic reaction is mediated by IgE class of antibodies. IgE binds with high affinity to Fc receptors on the surface of tissue mast cells and blood basophils. Such IgE coated mast cells and basophils are said to be ‘sensitized.’ A later exposure to the same allergen cross-links the membrane bound IgE on sensitized mast cells and basophils causing degranulation of these cells (Roit, 1988). The pharmacologically active mediators released from the granules exert biological effects like vasodilatation and smooth muscle contraction that may be either systemic or localized depending on the extent of mediator released. There are several probable mechanisms through which TPEIF could have prevented anaphylactic symptoms, when sensitized and subsequently challenged by BSA. TPEIF may act by provoking the production of IgG blocking antibodies, which binds the antigen (BSA) thereby preventing it from combining with IgE on the mast cell membrane.

It is convincing to assume that the whole extract of *Tridax procumbens* must be containing triterpenoides and sesquiterpene, which are reported responsible for the formation of nanoconstructs largely referred to as ISCOMS. These immunostimulatory complexes may presumably be held responsible for stimulation and potentiating of overall immune response. It is likely that some or the other protein/peptides of *Tridax procumbens* origin could had entrapped into these nanoconstructs. The later could have functionally presented them to the macrophages as well as to the cytosol components of dendritic cells (Antigen Presenting Cells). It is reported that antigen–iscom complex triggers immune response through secretion of IL-2 and IFN-γ, which are ultimately responsible for Th-1 and Th-2 cellular responses (Morein et al., 1993). Moreover, it is appropriate to realize that IFN-γ in specific, could be responsible for masking the IgE dependent anaphylactic shocks resulted from model immunogen, i.e. bovine serum albumin (Coffman and Carty, 1986). Other factors, which include specifically the sesquiterpene of Ca^{2+} ions, may be a major determinant in histamine release and its resultant affects (Foreman et al., 1977). It is relevant to recall that *Tridax procumbens* is reported some sesquiterpene and terpenoids, which are known to interplay a great role in overall immunological responses.

Using an animal model, we were able to measure the immunomodulatory effects of herbal treatment on the production of specific immunoglobulins following primary and secondary exposure to tetanus toxoid, used in clinical practice. To our knowledge, this is the first study that has explored the effect of immunomodulatory agent on the immune status of animal being challenged with a clinical antigen. This simulates the conditions of clinical vaccination and proposed the use of immunostimulant for a possible application in immunocompromised patients to generate sufficient immune response. All animals responded to the TT challenge and exhibited an enhanced IgG titer with booster dose on day 28. In comparison to control group, which received PBS (pH 7.4), TPEIF treated animals showed significant uplift of specific antibodies against TT.

In conclusion, the present study has shown the immunostimulatory activity of *Tridax procumbens* and suggests its therapeutic usefulness. TPEIF has stimulated both humoral as well as cellular arms of immune system. Further detailed studies are required which might establish a possible use of aqueous extract of *Tridax procumbens* Linn. in immunocompromised patients and as an adjuvant during vaccination programs in order to reduce number of non-responders to vaccines. However, detailed studies of mechanisms of immunomodulation and probable use in immunocompromised individuals are still to be investigated.

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References

Askenase, P.W., Van Loveren, M., 1983. Delayed type hypersensitivity: activation of mast cells by antigen specific T-cell factors initiates cascade of cellular interactions. Immunology Today 4, 259–264.

Atal, C.K., Sharma, M.L., Kaul, A., Khajuria, A., 1986. Immunomodulating agents of plant origin. 1. Preliminary screening. Journal of Ethnopharmacology 41, 185–192.

Coffman, R.L., Carty, J., 1986. A T-cell activity that enhances polyclonal IgE production and its inhibition by interferon-γ. Journal of Immunology 136, 949–954.

Diwan, P.V., Karwande, I., Margaret, I., Sattur, P.B., 1989. Pharmacology and biochemical evaluation of *Tridax procumbens* on inflammation. Indian Journal of Pharmacology 21, 1–7.

Estrada, A., Katselis, G.S., Laarveld, B., Barl, B., 2000. Isolation and evaluation of immunological adjuvant activities of saponins from *Polygala senega* L., comparative immunology. Microbiology and Infectious Diseases 23, 27–43.

Foreman, J.C., Hallet, M.B., Morgan, J.L., 1977. The relationship between histamine vsecretion and calcium++ uptake by mast cells. The Journal of Physiology (London) 271, 193–214.

Gonda, R., Tomoda, M., Shimizu, N., Kanari, M., 1990. Characterization of an acidic polysaccharide from the seeds of *Malva verticillata* stimulating the phagocytic activity of cells of the RES. Planta Medica 56, 73–76.

Hsu, H.C., Hsu, C.I., Lin, R.H., Kao, C.L., Lin, J.Y., 1997. Fip-vvo, a new fungal immunomodulatory protein isolated from *Volvariella volvacea*. Biochemical Journal 323, 557–565.

Jerne, N.K., Nordin, A.A., 1963. Plaque forming in agar by single antibody producing cells. Science 40, 405–407.

Kuby, J., 1994. Immunology, second ed. W.H. Freeman and Company, New York, pp. 23–45.

Lee, G.I., Young Ha, J., Rakmin, J., Nakagawa, H., Tsurufuji, S., Chang, I.M., Kim, Y., 1995. Inhibitory effects of oriental herbal medicines on IL-8 induction in lipopolysaccharide-activated rat macrophages. Planta Medica 61, 26–30.

Mediratta, P.K., Sharma, K.K., Singh, S., 2002. Evaluation of immunomodulatory potential of *Ocimum sanctum* seed oil and its possible mechanism of action. Journal of Ethnopharmacology 80, 15–20.

Morein, B., Villancres-Erikson, M., Akerblom, L., Lovquen, K., 1993. Initiation of immuno response with ISCOM. In: Gvegouardis, G., McCovmact, B., Adison, A.C., Porte, G. (Eds.), New Generation Vaccines: the Role of Basic Immunology, vol. 261. NATO ASI Series of Life Sciences, pp. 61–72.

Nelson, D.S., Mildenhall, P., 1967. Studies on cytophilic antibodies. 1. The production by mice of macrophage cytophilic antibodies to sheep erythrocytes: relationship to the production of other antibodies and development of delayed type hypersensitivity. Australian Journal of Experimental Biology and Medical Science 45, 113–130.

Pathak, A.K., Saraf, S., Dixit, V.K., 1991. Hepatoprotective activity of *Tridax procumbens*. Part I. Fitoterapia 62, 307–317.

Perumal, S.R., Ignacimuthu, S., Raja, D.P., 1999. Preliminary screening of ethnomedicinal plants from India. Journal of Ethnopharmacology 66, 235–240.

Picman, A.K., 1986. Biological activity of sesquiterpene lactones. Biochem. Syst. Ecol. 14, 255–281.

Puri, A., Saxena, R.P., Sumati-Guru, P.Y., 1992. Immunostimulant activity of picrlive, the iridoid fraction in mice. Planta Medica 58, 528–532.

Raghuvanshi, R.S., Singh, O., Panda, A.K., 2001. Formulation and characterization of immunoreactive tetanus toxoid biodegradable polymer particles. Drug Delivery 8, 99–106.

Ray, A., Banerjee, B.D., Sen, P., 1996. Modulation of humoral and cell-mediated immune responses by *Azadirachta indica* in mice. Indian Journal of Experimental Biology 34, 698–701.

Rezaeipoor, R., Saeidnia, S., Kamalinejad, M., 2000. The effect of *Plantago ovata* on humoral immune responses in experimental animals. Journal of Ethnopharmacology 72, 283–286.

Roit, I.M., 1988. Essential Immunology, sixth ed. ELBS Blackwell Scientific, Oxford, pp. 209–211.

SaiRam, M., Sharma, S.K., Ilavazhagan, G., Kumar, D., Selvamurthy, W., 1997. Immunomodulatory effects of NIM-76, a volatile fraction form Neem oil. Journal of Ethnopharmacology 55, 133–139.

Saraf, S., Pathak, A.K., Dixit, V.K., 1991. Hepatoprotective activity of *Tridax procumbens*. Part II. Fitoterapia 62, 534–536.

Sharma, M.L., Rao, C.S., Duda, P.L., 1994. Immunostimulatory activity of *Picrorhiza kurroa* leaf extract. Journal of Ethnopharmacology 41, 185–192.

Taddei, A., Rosas, R.A.J., 2000. Bioactivity studies of extracts from *Tridax procumbens*. Phytomedicine 7, 235–238.

Tzianabos, A.O., 2000. Polysaccharide immunomodulators as therapeutic agents: structural aspects and biological function. Clinical Microbiology Reviews 13, 523–533.

Udupa, S.L., Udupa, A.L., Kulkarni, D.R., 1991. Influence of *Tridax procumbens* on lysyl oxidase activity and wound healing. Planta Medica 57, 325–327.

Upadhyay, S.N., 1997. Plant products as immune response modulators. In: Proceedings of the International Ayurveda Conference-97. Sanjay Gandhi Post Graduate Institute of Medical Sciences, Lucknow, p. 10.

Wagner, H., 1983. Immunomodulatory agents. In: Proceedings of the Alfred Benzon Symposium, vol. 20. P. 559.